

STEP 1: CREATE DATABASE

```
-- Create database
CREATE DATABASE school;

-- Show databases
SHOW DATABASES;

-- Use database
USE school;
```

STEP 2: CREATE TABLE

```
-- Create students table
CREATE TABLE students (
    student_id INT,
    name VARCHAR(50),
    age INT,
    address VARCHAR(100)
);

-- View table structure
DESC students;
```

STEP 3: ALTER TABLE → ADD COLUMN

```
-- Add new column
ALTER TABLE students
ADD phone VARCHAR(15);

-- View updated structure
DESC students;
```

STEP 4: ALTER TABLE → MODIFY COLUMN

```
-- Modify size of name column
ALTER TABLE students
MODIFY name VARCHAR(100);

-- View updated structure
DESC students;
```

STEP 5: ALTER TABLE → RENAME COLUMN

```
-- Rename column name → student_name
ALTER TABLE students
CHANGE name student_name VARCHAR(100);

-- View updated structure
DESC students;
```

STEP 6: ALTER TABLE → DROP COLUMN

```
-- Remove phone column  
ALTER TABLE students  
DROP COLUMN phone;
```

```
-- View updated structure  
DESC students;
```

STEP 7: RENAME TABLE

```
-- Rename table  
RENAME TABLE students TO student_info;
```

```
-- View structure of renamed table  
DESC student_info;
```

⚠ From now onward, table name is student_info
NOT students

STEP 8: INSERT SINGLE RECORD

```
-- Insert single record  
INSERT INTO student_info  
VALUES (1, 'Ram', 18, 'Kathmandu');
```

```
-- View table data  
SELECT * FROM student_info;
```

STEP 9: INSERT MULTIPLE RECORDS

```
-- Insert multiple records  
INSERT INTO student_info  
VALUES  
(2, 'Sita', 19, 'Bhaktapur'),  
(3, 'Hari', 20, 'Lalitpur'),  
(4, 'Gita', 17, 'Pokhara');
```

```
-- View all records  
SELECT * FROM student_info;
```

STEP 10: SELECT QUERIES

✓ Select All Columns

```
SELECT * FROM student_info;
```

✓ Select Specific Column

△ Column name is now student_name
NOT name

```
SELECT student_name  
FROM student_info;
```

✓ Select Multiple Columns

```
SELECT student_name, age  
FROM student_info;
```

STEP 11: WHERE CLAUSE

✓ Age Greater Than 18

```
SELECT *  
FROM student_info  
WHERE age > 18;
```

✓ Multiple Condition

```
SELECT *  
FROM student_info  
WHERE age > 18  
AND address = 'Bhaktapur';
```

STEP 12: LIKE CLAUSE

✓ Starts With

```
SELECT *  
FROM student_info  
WHERE student_name LIKE 'R%';
```

✓ Ends With

```
SELECT *  
FROM student_info  
WHERE student_name LIKE '%a';
```

✓ Contains

```
SELECT *  
FROM student_info  
WHERE student_name LIKE '%it%';
```

STEP 13: UPDATE SINGLE COLUMN

```
-- Update address
UPDATE student_info
SET address = 'Chitwan'
WHERE student_id = 1;
```

```
-- View updated table
SELECT * FROM student_info;
```

STEP 14: UPDATE MULTIPLE COLUMNS

```
-- Update multiple columns
UPDATE student_info
SET age = 22,
    address = 'Butwal'
WHERE student_id = 2;
```

```
-- View updated table
SELECT * FROM student_info;
```

STEP 15: UPDATE ALL RECORDS

```
-- Increase age by 1
UPDATE student_info
SET age = age + 1;
```

```
-- View updated table
SELECT * FROM student_info;
```

STEP 16: DELETE SINGLE RECORD

```
-- Delete one student
DELETE FROM student_info
WHERE student_id = 4;
```

```
-- View remaining data
SELECT * FROM student_info;
```

STEP 17: DELETE MULTIPLE RECORDS

```
-- Delete students age less than 20
DELETE FROM student_info
WHERE age < 20;
```

```
-- View remaining data
SELECT * FROM student_info;
```

STEP 18: DELETE ALL RECORDS

```
-- Delete all records
DELETE FROM student_info;
```

```
-- View table
SELECT * FROM student_info;
```

STEP 19: TRUNCATE TABLE

```
-- Remove all records quickly
TRUNCATE TABLE student_info;
```

```
-- View empty table
SELECT * FROM student_info;
```

☞ Table structure remains.

STEP 20: DROP TABLE

```
-- Delete entire table
DROP TABLE student_info;
```

STEP 21: DROP DATABASE

```
-- Delete database
DROP DATABASE school;
```

🔥 What You Will See in MySQL

After DESC student_info;

You will see something like:

Field	Type	Null	Key
student_id	int	YES	
student_name	varchar(100)	YES	
age	int	YES	
address	varchar(100)	YES	

After SELECT * FROM student_info;

You will see:

student_id	student_name	age	address
1	Ram	18	Kathmandu
2	Sita	19	Bhaktapur

This Covers

DDL (Data Definition Language)

1. CREATE

- Create Database
- Create Table

2. ALTER

Requirement	Included
Add new column	ADD
Drop column	DROP COLUMN
Modify column	MODIFY
Rename column	CHANGE

3. DROP

Requirement
Drop table
Drop database

4. TRUNCATE

- Delete all records only
- Structure remains

5. RENAME

- Rename table

DML (Data Manipulation Language)

1. INSERT

Requirement
Single record
Multiple records

2. SELECT

Requirement
All columns
Specific column
Multiple columns

3. WHERE

Requirement
Single condition
Multiple condition (AND / OR)
Used in SELECT
Used in UPDATE
Used in DELETE

4. LIKE

Pattern
Starts with (R%)
Ends with (%a)
Contains (%ar%)
Wildcard (_)

5. UPDATE

Requirement
Single column update
Multiple column update
Update all rows
Uses WHERE properly

6. DELETE

Requirement
Single record
Multiple records
All records
Difference between DELETE vs TRUNCATE shown